

Problem 4

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Problem 1 A cup of coffee has temperature $20 + 75e^{-.02t}$ degrees (Celsius) t minutes after being poured into a cup. What is the average temperature of the coffee during the first half hour?

Explanation. Let $T(t) := 20 + 75e^{-.02t}$. We want the average value of T on $[0, 30]$.

$$\begin{aligned}
 T_{\text{avg}} &= \frac{1}{30 - 0} \int_0^{30} (20 + 75e^{-.02t}) \, dt \\
 &= \frac{1}{30} \left[20t - \frac{75}{.02} e^{-.02t} \right]_0^{30} \\
 &= \frac{1}{30} \left[20t - 3750e^{-.02t} \right]_0^{30} \\
 &= \frac{1}{30} [(20(30) - 3750e^{-0.6}) - (0 - 3750)] \\
 &= \frac{1}{30} (4350 - 3750e^{-0.6}) \\
 &= 145 - 125e^{-0.6}
 \end{aligned}$$

So the average temperature of the coffee during the first half hour is $145 - 125e^{-0.6} \approx 76.4$ degrees Celsius.